

# DOCUMENT 5 - MATHEMATICAL ROADMAP AND OPEN PROBLEMS

The 6D Projection Framework (Version 4.1)  
IFC Review Draft - Rev C

## Purpose

Version 4.1 establishes a conceptual architecture, epistemic boundary, and set of exploratory mechanisms. However, explanatory usefulness does not substitute for mathematical rigor.

This document lists the open mathematical and scientific problems that must be addressed for the framework to mature beyond conceptual synthesis.

The roadmap is not a proof of the framework. It is the punch list required to test whether the framework can survive formalization.

## Open Problem 1 - Projection Operators

The framework proposes that an underlying 3S+3T parent structure projects into an observable 3S+1T sector and a reciprocal 3T+1S sector.

The first mathematical task is to define explicit projection operators capable of carrying this architecture.

### Objective

Define projection operators  $Pi\_S$  and  $Pi\_T$  that map the parent structure into observer-accessible sectors while preserving internal consistency.

### Key Questions

- What is the mathematical structure of the parent space?
- Are the projections linear, nonlinear, gauge-selected, relational, or information-theoretic?
- What quantities are preserved under projection?
- What quantities are lost, hidden, or re-expressed as observer-accessible phenomena?
- Can the projection framework reproduce known 3S+1T physics as an effective limit?

## Open Problem 2 - Stabilization of Temporal Order

Version 4.1 explores the possibility that observable temporal order emerges through stabilization processes rather than existing as a pre-defined coordinate.

Candidate stabilization criteria include entropy, decoherence, record consistency, observer consistency, and causal stability.

### Objective

Develop a formal mathematical description of temporal stabilization.

### Exploratory Temporal Mapping Hypothesis

Several exploratory questions emerged during development of Version 4.1 concerning the relationship between temporal stabilization and observable temporal experience.

One possible organizational interpretation associates:

- X (Temporal) with Past,
- Z (Temporal) with Present,
- Y (Temporal) with Future.

and further explores possible correlations between:

- Decoherence and Past,
- Record Consistency and Present,
- Entropy and Future.

## Origin of the Z-Present Question

The proposal is not that the Present is a fundamental temporal coordinate. Rather, the Present may emerge along a preferred temporal-ordering direction selected by stabilization criteria.

By analogy with spatial gravity, which constrains preferred trajectories in physical space, a temporal stabilization process may constrain preferred trajectories within hidden temporal structure.

Under this interpretation, record consistency acts as a candidate selector for the active stabilization boundary experienced as Present.

## Temporal Selector Form

A possible formal expression is:

$$t_{\text{eff}} = n_i * \tau_i$$

where  $n_i$  represents a preferred ordering direction selected by stabilization criteria operating within the hidden temporal structure.

## Key Questions

- 1 How is a stable temporal sequence selected?
- 2 Does record consistency define a preferred ordering?
- 3 Can a temporal stabilization functional naturally generate Past, Present, and Future as organizational roles?
- 4 Are decoherence, record consistency, and entropy distinct mechanisms, or observer-accessible manifestations of a deeper stabilization process?
- 5 Can this mechanism explain the emergence of classical time without assuming time as fundamental?

## Open Problem 3 - Connectivity and Geometry

The framework explores the hypothesis that connectivity may organize geometry, from which gravity emerges.

### Objective

Develop a mathematical bridge between relational adjacency and observable geometric curvature.

### Candidate Structures

- relational adjacency matrices,
- graph-theoretic connectivity structures,

- information-theoretic mutual dependence,
- tensor-network-like relational geometries,
- emergent-metric constructions.

### Key Questions

- 1 Can geometric distance be derived from relational adjacency?
- 2 Can curvature emerge from connectivity density?
- 3 Can a relational network recover the tested limits of General Relativity?
- 4 Can an adjacency structure  $A_R(A,B)$  be connected to curvature quantities such as  $G_{\mu\nu}$ ?
- 5 Does gravitational attraction correspond to geometry responding to connectivity concentration?

## Open Problem 4 - Recovery of Quantum Mechanics

Any viable version of the framework must recover standard Quantum Mechanics where standard Quantum Mechanics is confirmed.

### Objective

Show whether the framework can reproduce standard quantum probability, superposition, interference, and entanglement as observer-accessible projection phenomena.

### Key Questions

- 1 Can the Born rule be recovered rather than assumed?
- 2 Can interference be represented as unresolved relational projection?
- 3 Can entanglement be formalized as relational adjacency without enabling faster-than-light signalling?
- 4 Can the framework reproduce Hilbert-space structure or explain why Hilbert-space structure appears?
- 5 Can decoherence emerge naturally from temporal stabilization?

## Open Problem 5 - Recovery of General Relativity

The framework must remain compatible with known gravitational physics.

### Objective

Determine whether General Relativity can arise as an effective observer-accessible limit of deeper relational organization.

### Key Questions

- 1 Can metric structure emerge from connectivity?
- 2 Can the Einstein field equations be recovered in an appropriate limit?
- 3 Can Lorentz consistency be preserved in the observable  $3S+1T$  sector?
- 4 Can gravitational lensing, orbital dynamics, and cosmological expansion remain compatible with known observations?
- 5 Can the framework avoid introducing unobserved violations of causality?

## Open Problem 6 - Dark Matter and Dark Energy

Version 4.1 explores whether some dark-sector phenomena may reflect inaccessible organizational structure rather than only unseen matter-energy content.

### Objective

Identify whether the framework can produce testable consequences relevant to dark matter or dark energy phenomena.

### Key Questions

- 1 Can hidden connectivity density mimic gravitational effects currently attributed to dark matter?
- 2 Can large-scale reorganization or dilution of connectivity produce expansion-like geometric effects?
- 3 Can the framework distinguish itself from particle dark matter, modified gravity, or cosmological constant models?
- 4 Can it generate predictions for galaxy rotation curves, gravitational lensing, structure formation, or expansion history?
- 5 What observations would falsify the dark-sector interpretation?

## Open Problem 7 - Toy Models

Before a full physical theory can be built, simplified models are required.

### Objective

Construct toy models showing how relational structure can project into observable order.

### Candidate Toy Models

- dependency-network to calendar projection,
- graph adjacency to spatial distance,
- temporal stabilization selector models,
- decoherence-record consistency models,
- simple two-sector projection systems.

### Success Criteria

- The toy model must be explicit.
- It must define its variables.
- It must generate observable structure from underlying relationships.
- It must show what is preserved and what is lost under projection.
- It must identify where the analogy breaks down.

## Open Problem 8 - Falsifiability and Failure Conditions

A scientific framework must identify not only what would support it, but also what would count against it.

### Objective

Define clear failure conditions for Version 4.1.

### **Potential Failure Conditions**

- Failure to recover standard Quantum Mechanics in confirmed domains.
- Failure to preserve Lorentz causality.
- Failure to define projection operators coherently.
- Failure to distinguish the framework from existing interpretations.
- Failure to produce any discriminating mathematical or observational consequence.
- Collapse into unfalsifiable metaphor rather than formal structure.

## **Open Problem 9 - Recovery of Existing Physics and Comparative Mapping**

The framework must be compared against existing theories and research programs rather than presented in isolation.

### **Objective**

Identify where Version 4.1 is equivalent to existing work, where it is interpretive, where it provides a useful synthesis, and where it makes distinct claims.

### **Conceptual and Structural Benchmarking**

Future mathematical development should explicitly compare the framework against Two-Time Physics, Relational Quantum Mechanics, Quantum Graphity, decoherence/einselection, thermodynamic arrow-of-time literature, and information-theoretic observer frameworks.

The purpose is to identify:

- where V4.1 is equivalent to existing work,
- where it is merely interpretive,
- where it provides a useful synthesis,
- and where it makes genuinely distinct claims.

### **Key Questions**

- 1 What is already covered by Two-Time Physics or other higher-dimensional projection programs?
- 2 What is already covered by Relational Quantum Mechanics or QBism-like observer frameworks?
- 3 What is already covered by Quantum Graphity, tensor networks, or emergent-geometry models?
- 4 What is already covered by decoherence/einselection and thermodynamic arrow-of-time literature?
- 5 What remains genuinely distinct in Version 4.1 after comparison?

## **Open Problem 10 - Mathematical Status of the Integration Principle**

The central organizing principle of Version 4.1 is:

## **Relationship → Organization → Observable Order**

### **Objective**

Determine whether this principle can be formalized as mathematics rather than remaining a guiding intuition.

### **Key Questions**

- 1 Can relationship be represented as a primitive mathematical structure?
- 2 Can organization be defined as a transformation or stabilization process?
- 3 Can observable order be derived as an output of that process?
- 4 Can this principle be connected to known structures such as category theory, graph theory, information theory, tensor networks, or variational principles?
- 5 Can the principle generate quantitative predictions?

### **Current Assessment**

At the present stage, Version 4.1 is strongest as a conceptual integration framework.

Its strongest elements are:

- the Embedded Observer Problem,
- Relationship → Organization → Observable Order,
- the Temporal Stabilization Hypothesis,
- the connectivity-geometry research direction,
- and the explicit distinction between hypothesis, mechanism, analogy, and proof.

Its weakest elements are the areas where mathematical formalization remains missing: projection operators, stabilization functionals, metric emergence, recovery of known physics, and testable predictions.

### **Success Criterion**

The framework does not need to prove every speculative element immediately to be useful.

It succeeds at the next stage if it can generate one or more mathematically explicit toy models showing how observable order emerges from relational organization.

It matures scientifically only if those toy models can eventually recover known physics or produce discriminating predictions.

### **Closing Note**

This roadmap is not a defense of the framework.

It is a list of ways to test it.

The aim is not belief.

The aim is interrogation.